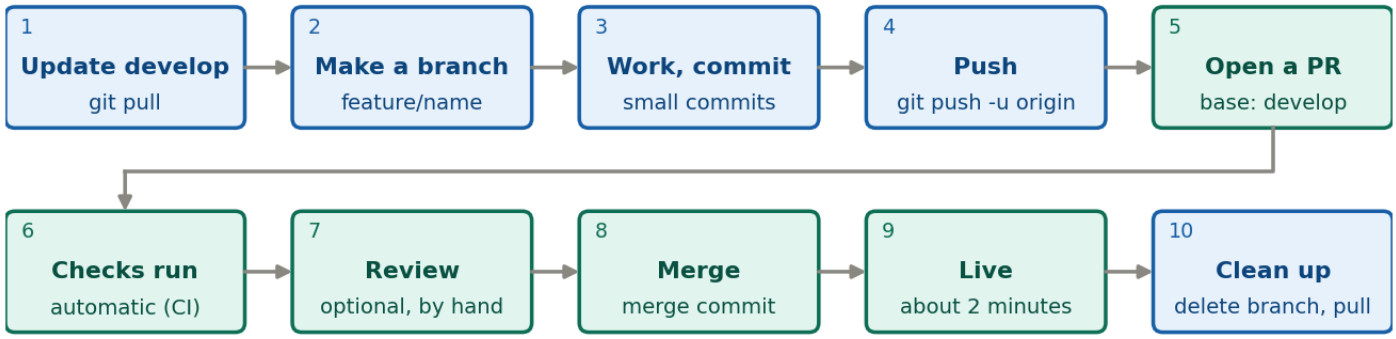
Developer workflow

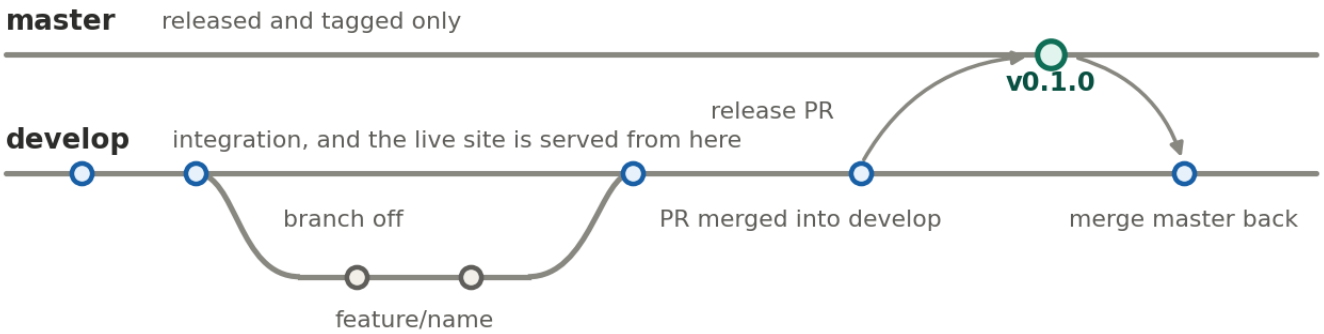
NSDF-Data · Git Flow and semantic versioning · version 4, 23 October 2026

A pull request (PR) is a page on GitHub that says "please merge branch X into branch Y". It shows the changes, runs the automatic checks, and keeps the discussion in one place. **Nothing reaches** `develop` **until you press Merge.** `develop` is where work lands and what the live site is served from; `master` holds tagged releases only.



The loop for every change. Blue: on your computer. Teal: on GitHub. A check turns red? Fix it on the same branch and push again; the PR updates.

```
git switch develop && git pull                          # 1  start from the latest develop
git switch -c feature/my-change                         # 2  your branch
git add <files> && git commit -m "What and why"         # 3  small commits; update CHANGELOG.md
git push -u origin feature/my-change                    # 4
gh pr create --base develop --fill                      # 5  or "Compare & pull request" on GitHub
# 6-9: checks run, optional review, press Merge (merge commit), the site updates
git switch develop && git pull && git branch -d feature/my-change   # 10
```



Branches over time. A feature merges into `develop` through a PR. A release goes to `master`, is tagged, and `master` is merged back.

Making a release, when there is something to version: branch `release/X.Y.Z` from `develop`, set `VERSION` (and the library's own version in `python/pyproject.toml` if the library changed) and date `CHANGELOG.md`, open a PR into `master` and merge it, tag `vX.Y.Z` on the merge commit, merge `master` back into `develop`, publish the GitHub release, delete the branch. Bump **major** for a breaking library change (not while 0.x), **minor** for a new capability, **patch** for fixes and content. A guide or site fix is an ordinary feature PR. A **hotfix** is only for a bug in a released version.

Special cases. *A student note:* the PR into `develop` runs the check only, and nothing is visible; you replace `TBD` with the next note number (`python tools/build_notes.py next` prints it) and merge, the bot builds the page, and it is live in 2 to 3 minutes (delete the source file to remove a note). *A contributor such as Kitty:* they open a PR into `develop`, you are requested automatically, and you add other reviewers by hand. *An AI agent:* may open PRs, and never merges without your go-ahead.

Never commit directly to `master` or `develop`, and never force-push (the rulesets block both).

Details are in the repository: `RELEASING.md`, `AGENTS.md`, `CHANGELOG.md`.